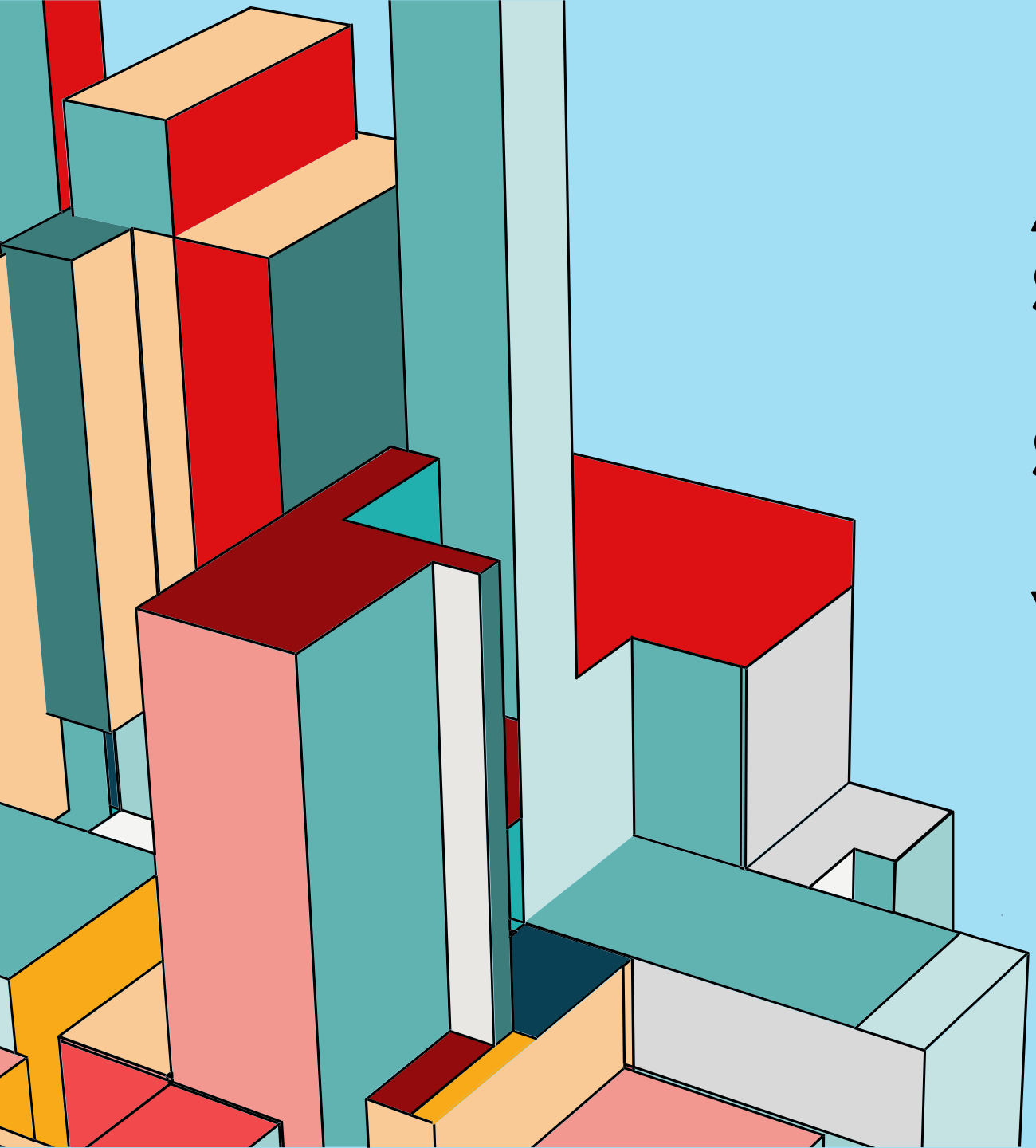


APPLIED DATA SCIENCE PROJECT

SHOT RECOGNITION IN PADEL MATCHES VIDEOS

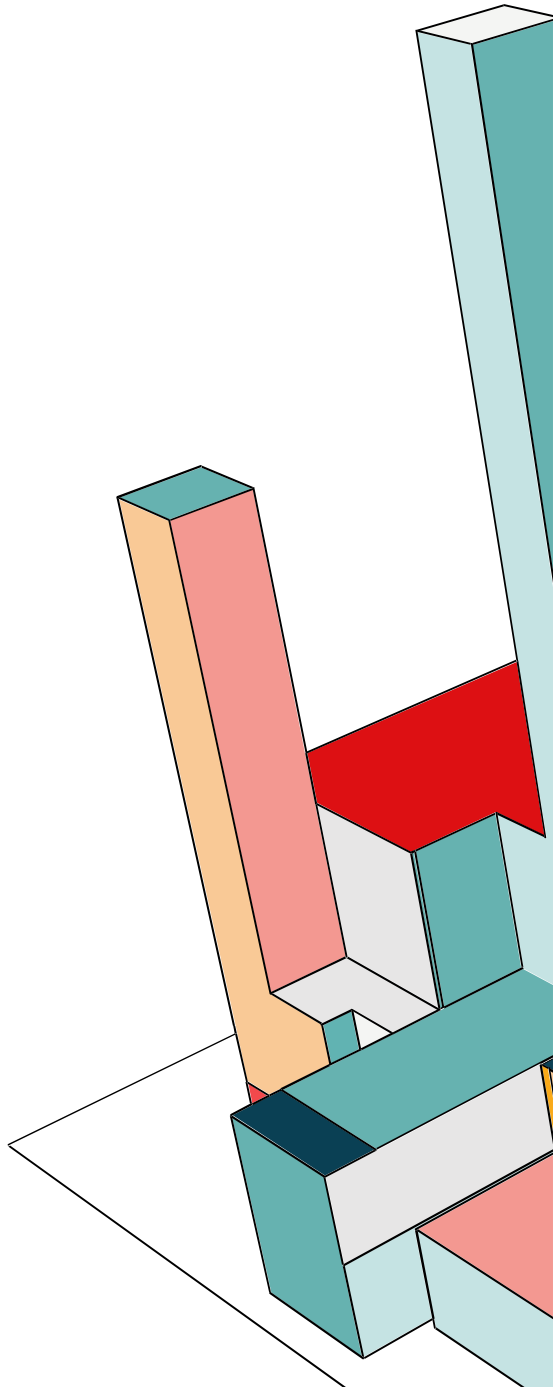
Politecnico di Torino

FONDAZIONE links PASSION FOR INNOVATION

ellis European Laboratory for Learning and Intelligent Systems

VALUE-DRIVEN PROJECT

- Automatic **analysis** of **sports videos** has gained increasing attention in recent years, driven by its potential applications in performance analysis and tactical evaluation
- A previous project addressed the **localisation of ball and players**: recognising **which shots** are performed, and **when**, is the natural next step to extract meaningful statistics from padel matches
- This project aims to develop a machine learning algorithm to automatically **detect and classify the shots** performed by the players in a padel match video



DATA

- Padel Tracker 100 Dataset:

- Two extracts from 2022 World Padel Tour Finals



- With annotations of

Ball position

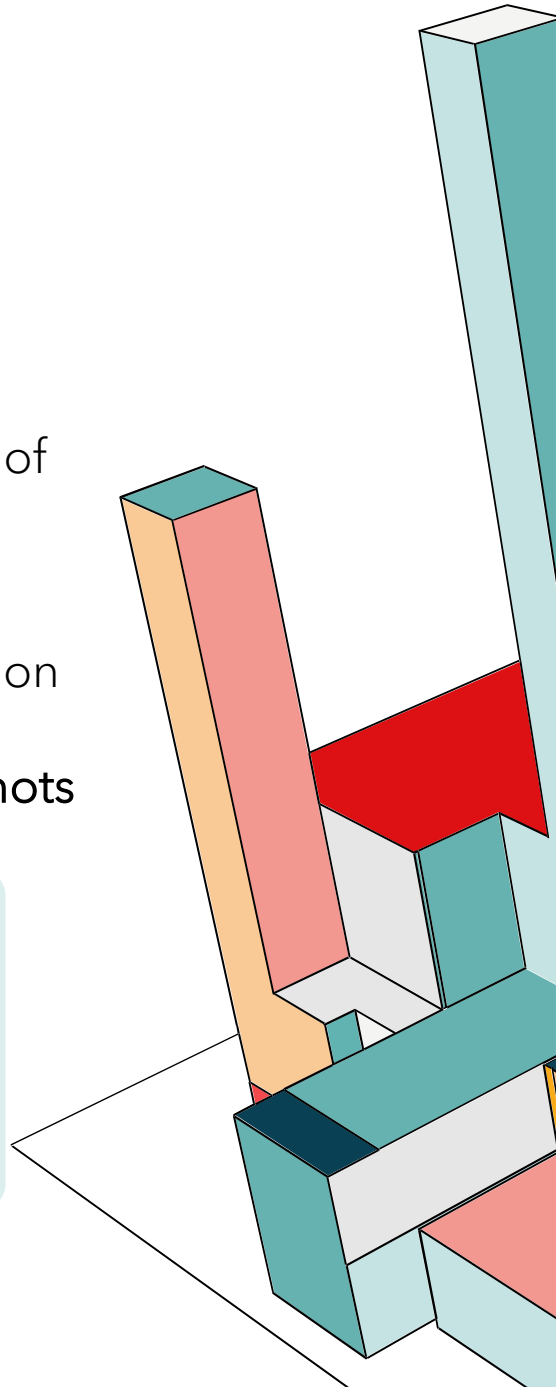
Players position

Performed shots

Target: shot events

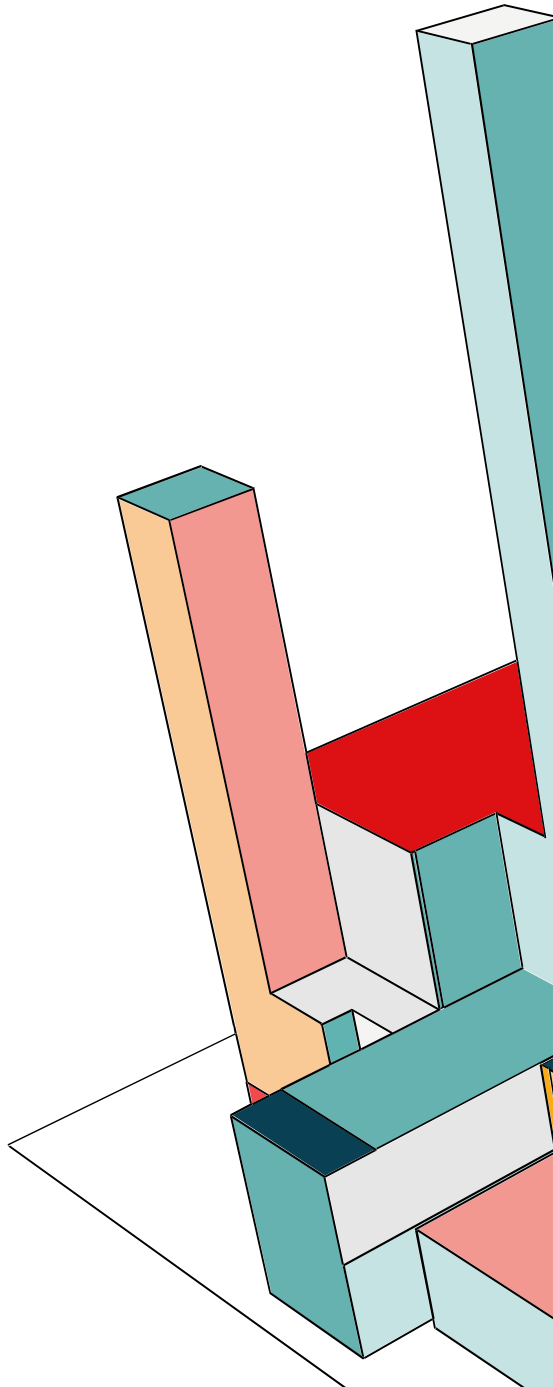
6 classes: forehand, backhand, smash, serve, dropshot, other

~40k annotated frames, plus player poses



TASK

- **Shot Recognition** task where the objective is to detect **when** a shot is performed in the video and **which type** of shot it is
- Participants are thus tasked to develop a **machine-learning** model that processes **sequences of frames** of input video data and outputs the shot events with their class
- Besides raw frames, the model may exploit ball trajectory, player positions and **poses**
- **No existing baseline**: the evaluation metrics will be defined within the project and the results obtained will become the first reference



METHODOLOGY AND WORK PLAN

- Since no previous work exists for this task, the project starts with a **research phase** to select the models to adopt

1

State of the art

Review of models for action recognition and event spotting in sports videos (frame-, pose- and trajectory-based)

2

Data analysis

Exploration of shot annotations, class distribution and preparation of training clips

3

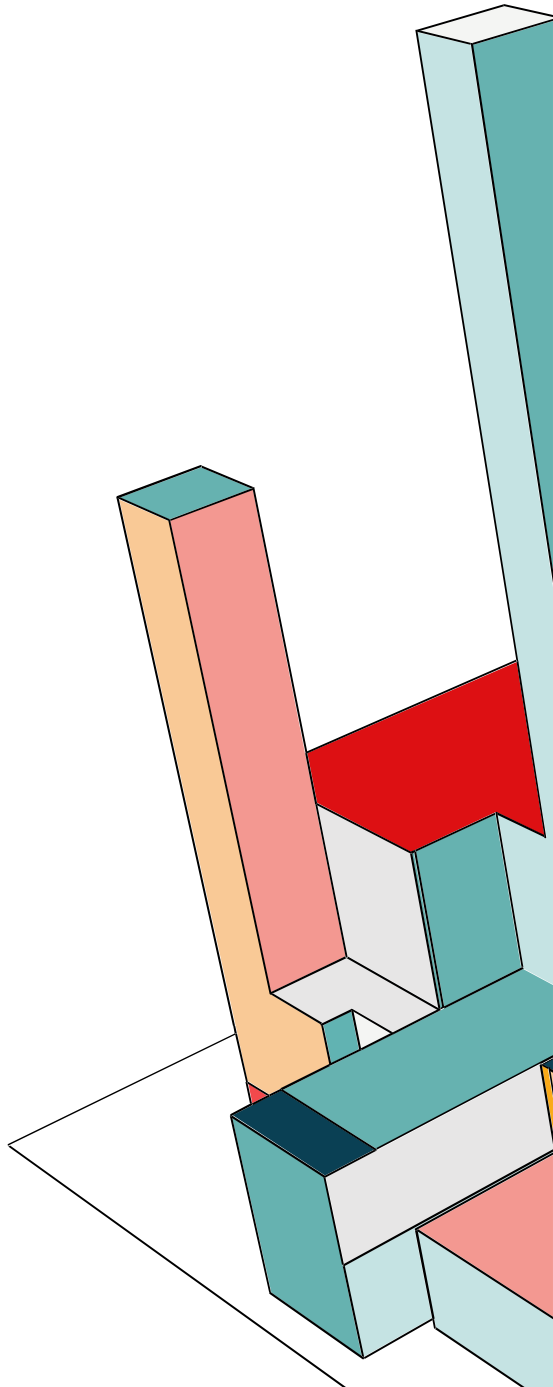
Model development

Implementation and training of the selected candidate models

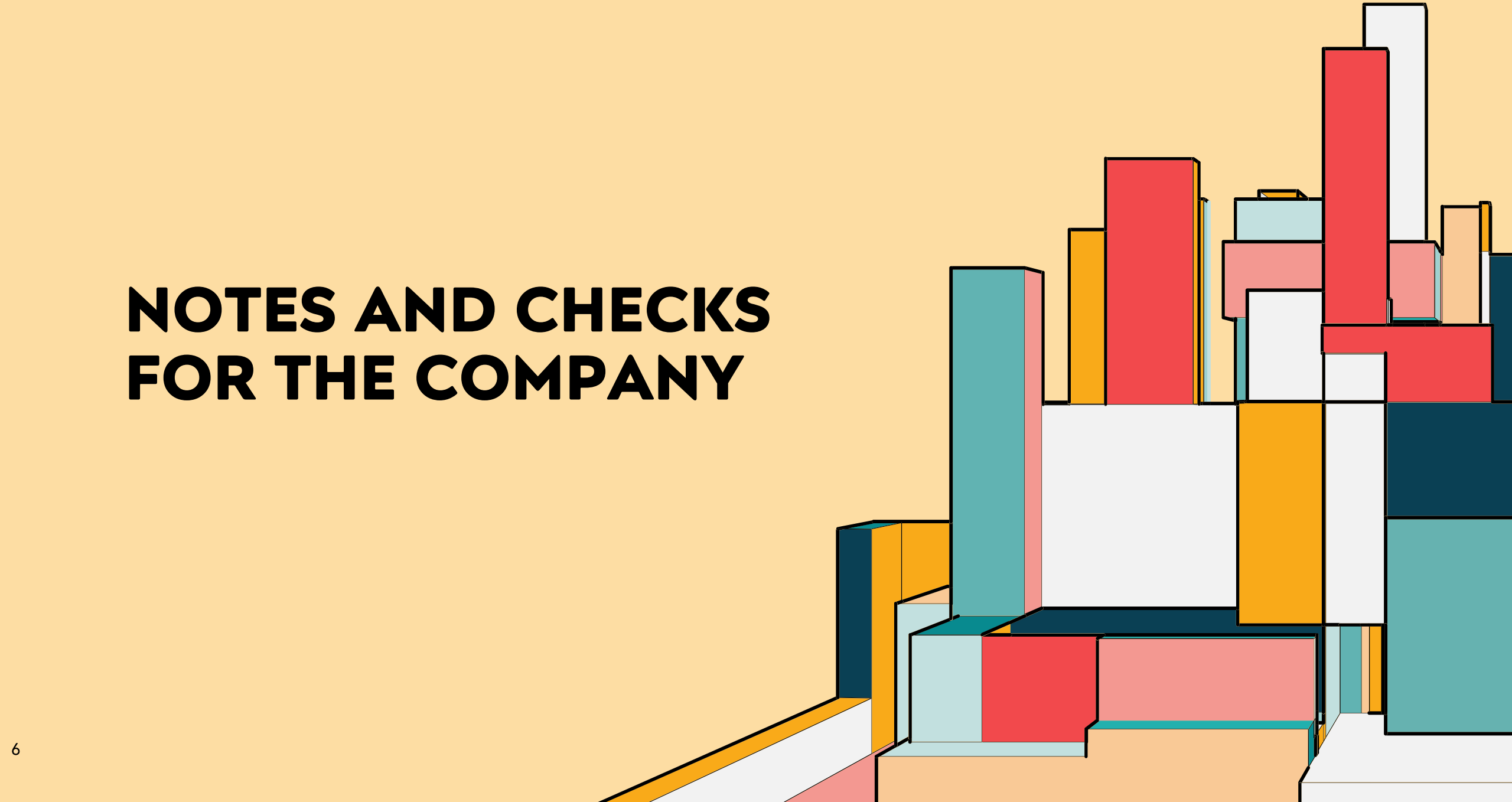
4

Evaluation

Definition of metrics, comparison of models and first reference results

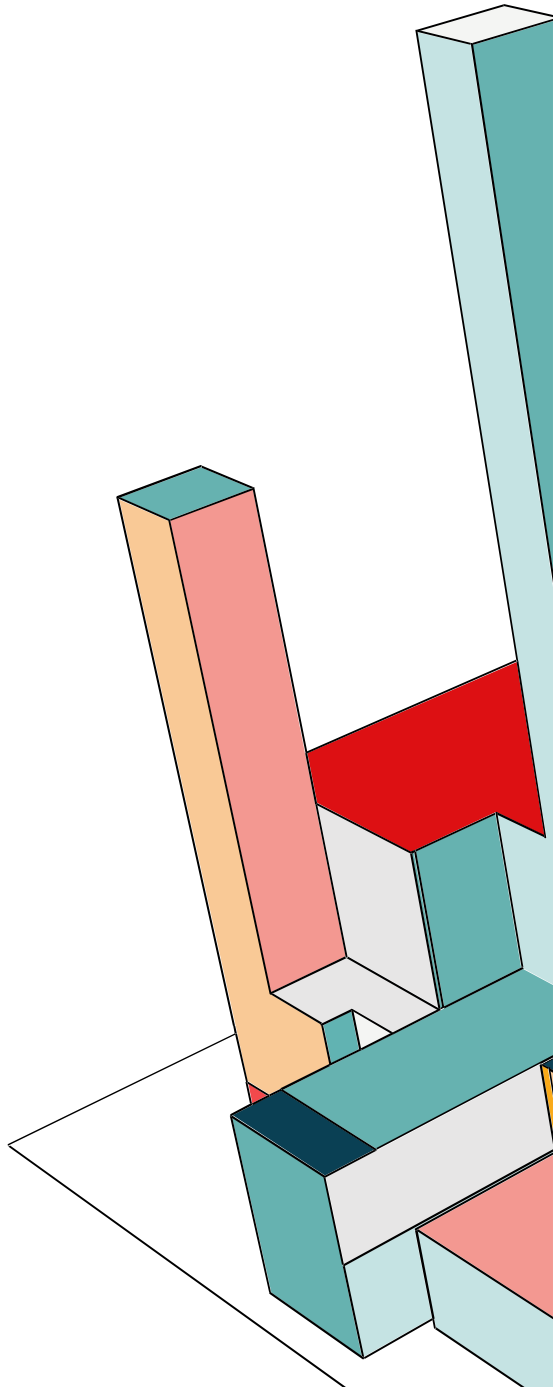


NOTES AND CHECKS FOR THE COMPANY



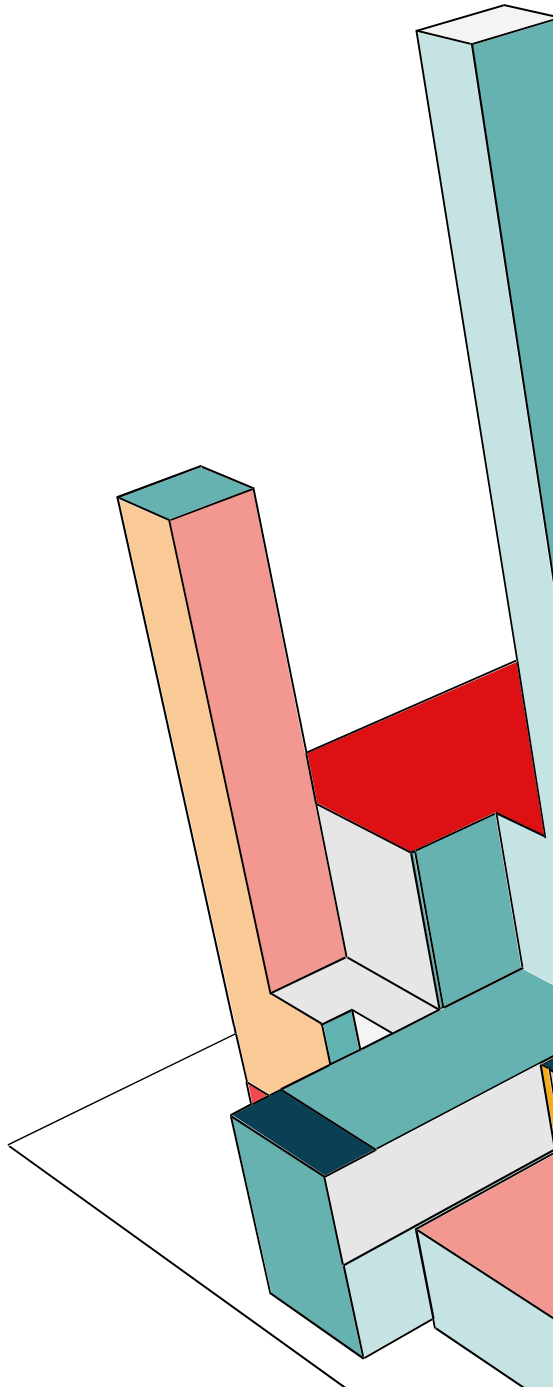
LIGHT MENTORING

- The following meetings will be organized:
  - **First (longer) meeting** as introduction to the project context, objectives and data presentation.
  - **Periodic bi-weekly meetings** (1h) with LINKS to align with the current development.



POLICY

- The Padel Tracker 100 dataset is publicly available under a CC BY 4.0 license
- No NDA is required: data can be freely used, provided the dataset authors are credited
- Code and methodology could be published



THANKS FOR THE ATTENTION

CONTACTS

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